Calibrated Multi-Scale Similarity Fusion for Wildlife Individual Re-Identification on WildlifeReID-10k

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1 Introduction

Wildlife individual re-identification (Re-ID) is the task of determining whether different images show the same individual animal. Unlike species classification, which asks what type of animal appears in an image, Re-ID asks which particular individual is shown. This distinction is important for wildlife monitoring because the same individual may appear differently across photographs due to changes in viewpoint, lighting, pose, body condition, or environment, while different individuals of the same species may have very similar appearances. Reliable individual identification is therefore a key requirement for long-term monitoring without physically tagging or repeatedly handling animals [14].

Traditional approaches to monitoring individual animals often rely on physical tags, collars, or other forms of identification. Although effective in some settings, these methods can be invasive, expensive, and difficult to apply consistently across many species and locations. Image-based Re-ID provides a less invasive alternative: once an individual has been photographed, later images can potentially be matched to that individual without additional capture or handling. This can support conservation applications such as population monitoring, estimating animal abundance, and studying changes in individuals over time [7].

However, reliable wildlife Re-ID remains difficult in unconstrained photographic collections. Images may contain substantial variation in background, viewpoint, illumination, image quality, and animal pose. In addition, individuals of the same species may have only subtle visual differences. These challenges become more pronounced in large heterogeneous benchmarks that combine images from different datasets, locations, and species. A representation that performs well on one collection may therefore fail to generalise to another.

One possible way of addressing this problem is to use complementary visual representations rather than relying on a single descriptor. A global representation can capture broad characteristics such as body shape and morphology,

while local features can capture fine-grained patterns such as spots, stripes, or other distinctive markings. A third, general-purpose visual representation may provide additional information that is not fully captured by either global morphology or local correspondences. The combination of these representations may therefore provide a more complete description of an individual animal than any one representation alone. Recent work on calibrated fusion has shown that combining global and local evidence can improve wildlife Re-ID performance [7], while more recent approaches have explored adding a third visual representation [**?**]. This project investigates whether such complementary representations can provide consistent benefits at the scale and diversity of WildlifeReID-10k, which contains 140,488 images of 10,772 individuals across 37 source datasets and roughly 33 species [2]. The project focuses on calibrated fusion of three types of visual evidence: a global wildlife individual descriptor, a complementary visual representation, and local keypoint-based evidence. MiewID and a fine-tuned MegaDescriptor are considered as alternative global representations, while EVA02 provides the complementary representation and ALIKED provides local matching evidence. Their contributions are evaluated individually and in combination to determine whether the representations contain complementary information for individual level retrieval.

The project also considers whether the resulting fused representation remains effective when the system encounters individuals that were not observed during training. This is particularly important for wildlife monitoring because real populations are not fixed: new individuals may enter a monitored population, while previously observed individuals may disappear. The study therefore evaluates both closed-set and open-set Re-ID using the evaluation protocols provided by WildlifeReID-10k. In addition, because combining several representations increases the dimensionality and storage requirements of the resulting representation, the project investigates how much it can be compressed before individual-level retrieval performance begins to degrade. The main research objective is therefore to determine whether calibrated fusion of global, complementary, and local visual representations provides a more effective basis for wildlife individual Re-ID than individual representations alone, and to establish how well this approach generalises across the heterogeneous WildlifeReID-10k benchmark. The study further investigates the behaviour of the fused representation under open-set conditions and the extent to which it can be compressed while retaining individual discriminative information.

The remainder of this proposal is organised as follows. Section 2 describes the problem addressed by the project and defines the research questions. Section 3 reviews the related work and identifies the research gap. Section 4 presents the proposed methodology and evaluation strategy. Section 5 justifies the proposed approach and summarises the expected contribution.

2 Problem Description

Wildlife individual re-identification becomes increasingly difficult as the number and diversity of images and individuals increase. WildlifeReID-10k provides a suitable setting for studying this problem because it combines 140,488 images of 10,772 individuals from 37 source datasets and approximately 33 species [2]. Unlike a small, single-species collection, this benchmark contains substantial variation in animal appearance, image quality, photographic conditions, and the number of images available for each individual. A method that performs well on one subset of the data may therefore not necessarily generalise to the wider benchmark.

A major difficulty is learning individual-specific information without relying on incidental visual information. Wildlife images are often captured in natural environments where backgrounds, lighting, vegetation, terrain, and camera viewpoints vary considerably. If a representation relies too heavily on these factors, it may associate an individual with the environment in which it was photographed rather than with characteristics of the animal itself. Animal detection and cropping can reduce this risk by focusing the subsequent representation on the animal, but the resulting representation must still retain sufficiently detailed information to distinguish individuals that may have very similar appearances.

This difficulty is closely related to the limitations of relying on a single visual representation. Global descriptors can capture broad characteristics such as body shape and morphology, but may not fully represent small, spatially localised differences between individuals. Local matching methods can provide evidence from distinctive patterns such as spots, stripes, or scars, but their matching behaviour differs from that of fixed-length global embeddings. A general-purpose visual representation may provide additional information that is not captured by either global morphology or local correspondences. It is therefore necessary to determine whether these different forms of visual evidence contain complementary information that can be combined effectively, rather than assuming that a single representation is sufficient for individual-level identification.

The problem is further complicated when the system encounters individuals that were not observed during training. In a conventional closed-set setting, the identities evaluated at test time are assumed to be represented in the training data. The system can therefore select the most similar known identity from a fixed set of candidates. In an open-set setting, this assumption is relaxed: a query may belong to an individual that was not observed during training. The system must consequently distinguish between a sufficiently strong match to a known individual and a query for which no known identity should be assigned. This setting is particularly relevant to wildlife monitoring because populations are not static and new individuals may appear over time.

These settings can be expressed formally. In the closed-set case, the test identities are contained within the identities available during training,

$$ID_{\text{test}} \subseteq ID_{\text{train}}.$$

In the open-set case, at least some evaluated identities are absent from the training identities,

$$ID_{\text{test}} \not\subseteq ID_{\text{train}}.$$

The open-set problem therefore requires not only ranking candidate identities but also determining whether the similarity to the best candidate is sufficiently strong to accept that identity.

WildlifeReID-10k also presents substantial variation in the number of images available for different individuals. Some individuals are represented by many images, while others have very few observations. A single pooled accuracy value may therefore provide an incomplete picture of performance because frequently observed individuals can contribute disproportionately to the overall result. The benchmark's evaluation protocols and BAKS and BAUS metrics provide a way of assessing performance while accounting for this variation [2, 11]. Using these benchmark metrics also allows the proposed approach to be evaluated under the same conditions as other methods developed for WildlifeReID-10k.

Taken together, these factors motivate the central problem addressed by this project: *whether multiple complementary visual representations can be calibrated and fused to provide more reliable individual-level wildlife Re-ID than any single representation alone, while remaining effective for unseen identities and across the heterogeneous data represented in WildlifeReID-10k.*

The project consequently investigates four related questions: (i) whether MiewID or a fine-tuned MegaDescriptor provides the stronger global representation on WildlifeReID-10k; (ii) whether combining the selected global representation with EVA02 and ALIKED provides additional individual-discriminative information compared with the individual representations alone; (iii) whether the resulting fused representation remains effective under the benchmark's open-set evaluation, where some identities are not observed during training; and (iv) how much the fused representation can be compressed before its individual-level retrieval performance begins to degrade.

These questions define the scope of the research. The core investigation is the representation and evaluation of calibrated multi-scale feature fusion. The research pipeline therefore focuses only on the core research stages, from visual representation and score calibration through fusion, compression, retrieval, and evaluation.

3 Related Work

Wildlife individual re-identification has developed from species-specific identification systems towards more general multi-species representations. This development has been driven by the increasing availability of wildlife imagery and by advances in large-scale visual representation learning. WildlifeReID-10k reflects this development by bringing together images from 37 source datasets and providing both closed-set and open-set evaluation protocols designed to reduce information leakage between training and testing data [2]. The following work is therefore relevant to this project because it establishes the progression from

general-purpose visual representations to wildlife-specific descriptors and, more recently, to calibrated fusion of multiple complementary representations.

General-purpose visual representations. Self-supervised vision transformers provide a useful starting point for learning visual representations without requiring task-specific labels. DINO [6] introduced a self-distillation framework in which different augmented views of an image are encouraged to produce consistent representations. DINOv2 [12] extended this approach through larger-scale and more carefully curated training data, producing general-purpose visual features that transfer across many visual tasks. In wildlife Re-ID, such general-purpose representations provide useful baselines because they can encode visual information without having been trained specifically to identify individual animals. However, their general visual training does not directly optimise the fine-grained individual-specific distinctions required by wildlife Re-ID. This motivates the development and comparison of representations trained specifically for animal identification.

Wildlife-specific global representations. WildlifeDatasets and the associated MegaDescriptor model represent an important step towards general-purpose wildlife Re-ID. Cermak et al. [**?**] introduced WildlifeDatasets as an open-source framework for organising wildlife Re-ID datasets and training models across multiple species. MegaDescriptor was developed as a wildlife-specific foundation model for individual animal Re-ID, providing a strong baseline against which more specialised approaches can be compared. The WildlifeReID-10k benchmark also uses MegaDescriptor as a baseline, reinforcing its relevance to large-scale multi-species evaluation. The benchmark itself was introduced partly to address the limitations of evaluating wildlife Re-ID on small, isolated datasets and to provide similarity-aware splits that reduce leakage from visually similar images appearing across training and test sets [2].

MiewID provides another wildlife-specific global representation. Otarashvili et al. [**?**] proposed MiewID as a multi-species individual identification model based on an EfficientNetV2 backbone and an angular-margin training objective. Its multi-species training and reported generalisation to species outside its training data make it a relevant alternative to MegaDescriptor. Since MiewID and MegaDescriptor provide different approaches to learning global individual-level representations, there is no need to assume beforehand that one is universally superior. This project therefore compares them on the same WildlifeReID-10k evaluation setting before selecting the global representation used in the proposed fusion.

Global and local feature fusion. Although wildlife-specific global descriptors provide strong individual representations, they may not capture every fine-grained visual difference between two individuals. Local feature matching provides a complementary source of evidence by comparing spatially localised patterns between images. WildFusion [7] addresses this problem by combining global similarity with local keypoint-based matching. Rather than directly averaging raw scores, WildFusion calibrates the different similarity measures before combining them, allowing global and local evidence to contribute on a common scale.

This calibration step is particularly important because global embedding similarities and local matching scores have different numerical distributions. WildFusion therefore provides the methodological basis for the fusion strategy investigated in this project. Its results demonstrate that calibrated global-local fusion can improve wildlife Re-ID performance over individual global representations, supporting the hypothesis that different visual representations can contain complementary identity information [7].

Multi-representation fusion. More recent work has extended global-local fusion by introducing a third visual representation. Kim et al. [**?**] combined MegaDescriptor, ALIKED, and EVA02 for the AnimalCLEF 2025 individual animal Re-ID challenge. MegaDescriptor provided global visual features, ALIKED provided local keypoint-based matching, and EVA02 supplied an additional visual representation. The three components were combined using WildFusion-style score calibration followed by weighted fusion. This work provides direct motivation for the three-representation configuration investigated in this project.

However, the role of the third representation should not be assumed to be beneficial simply because it improves performance in one challenge setting. The important research question is whether the additional representation continues to provide complementary information when evaluated on a larger and more heterogeneous benchmark. In particular, it remains necessary to compare the fused system against its individual components under a common evaluation protocol and determine whether the observed improvement is consistent across the different species and source datasets represented in WildlifeReID-10k.

Open-set wildlife re-identification. Open-set recognition is particularly important for wildlife Re-ID because a deployed system cannot assume that every future observation belongs to an individual previously registered in its database. WildlifeReID-10k explicitly addresses this issue by providing open-set evaluation in addition to closed-set evaluation. Its similarity-aware split was introduced to reduce the effect of visually similar images appearing in both training and testing data, which can otherwise lead to overly optimistic estimates of Re-ID performance [2].

The importance of open-set evaluation is also reflected in AnimalCLEF. AnimalCLEF 2025 uses an open-set formulation in which the system must determine both whether the query individual exists in the database and, when it does, which identity it corresponds to [**?**]. These benchmarks demonstrate that wildlife Re-ID should not be evaluated only as closed-set classification. A representation intended for practical monitoring must also be able to reject sufficiently dissimilar queries rather than assigning every query to a known identity.

Balanced evaluation across individuals. Wildlife datasets are often highly imbalanced because some individuals are photographed much more frequently than others. This creates a problem for standard pooled accuracy measures, which can allow frequently observed individuals to dominate the evaluation. WildlifeReID-10k therefore provides balanced evaluation measures for known and unknown identities, including BAKS and BAUS, to provide a more informative assessment under its open-set setting [2]. These metrics are particularly relevant to

this project because the objective is not merely to improve performance on frequently represented individuals, but to determine whether the fused representation remains useful across the diverse identities represented in the benchmark.

Representation compression and similarity retrieval. A multi-representation approach increases the amount of information that must be stored and compared for each image. This introduces a secondary practical question: how much of the resulting representation is necessary to retain its individual-discriminative information? Principal Component Analysis (PCA) [**?**] provides a standard approach for reducing the dimensionality of high-dimensional representations while retaining directions of greatest variance. However, the directions with highest overall variance are not necessarily those that best distinguish individual animals. Compression must therefore be evaluated empirically rather than assumed to preserve individual identity information.

Similarity retrieval also becomes increasingly expensive as the gallery grows. Approximate nearest-neighbour methods such as locality-sensitive hashing (LSH) [**?**] can reduce the number of comparisons required during retrieval. In this project, however, retrieval efficiency is treated as a supporting consideration rather than the primary research objective. The main question is whether compression can reduce the representation size while retaining individual-level retrieval performance.

Resulting research gap. The existing literature establishes several important foundations for this project. General-purpose representations such as DINOv2 provide strong visual baselines, while wildlife-specific models such as MegaDescriptor and MiewID are designed specifically for individual animal identification. WildFusion demonstrates that calibrated fusion of global and local evidence can improve wildlife Re-ID, and the AnimalCLEF 2025 work demonstrates that a third representation, EVA02, can be incorporated into a similar fusion framework [**?**,7, **?**,**?**].

What remains less clear is whether this multi-representation approach provides consistent individual-level benefits when evaluated at the full scale and diversity of WildlifeReID-10k. In particular, it is not yet clear whether the complementary information from global, local, and additional visual representations remains useful across the benchmark's different species and source datasets, whether the resulting representation remains effective when identities are unseen during training, and how much of the fused representation can be removed through dimensionality reduction before individual-level discrimination begins to deteriorate. These questions motivate the proposed evaluation of calibrated fusion using MiewID or fine-tuned MegaDescriptor as the global representation, EVA02 as a complementary visual representation, and ALIKED as the local representation under WildlifeReID-10k's benchmark protocols.

4 Proposed Approach

Figure 1 presents the proposed research methodology. The methodology focuses only on the core research process used to investigate multi-scale feature fusion for

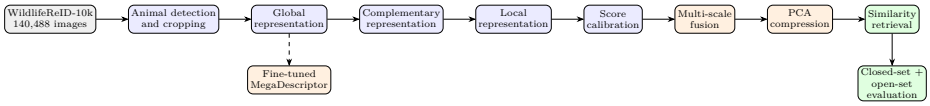


Fig. 1. Core research methodology for multi-scale calibrated feature fusion on WildlifeReID-10k. Images are first processed using animal detection and cropping. Global, complementary, and local representations are then extracted and calibrated before being combined through multi-scale fusion. The resulting representation is evaluated under different compression levels and used for similarity-based retrieval under closed-set and open-set protocols.

wildlife individual re-identification. WildlifeReID-10k images are first prepared through animal detection and cropping. The cropped images are then processed using complementary visual representations: a global wildlife-specific descriptor, a general visual descriptor, and a local keypoint-based descriptor. Their similarity scores are calibrated before being combined through weighted fusion. The resulting fused representation is subsequently evaluated for compression, similarity retrieval, and closed-set and open-set re-identification. The methodology is designed to determine whether combining different forms of visual evidence provides more reliable individual-level discrimination than using a single representation.

Dataset and evaluation setting. The project uses WildlifeReID-10k [2] as the primary evaluation benchmark. The benchmark contains 140,488 images from 37 source datasets, covering approximately 33 species and 10,772 individual animals. The experiments use the benchmark's published evaluation protocols rather than creating an alternative splitting strategy, allowing the results to be compared with existing work on the same benchmark.

The evaluation considers both closed-set and open-set settings. In the closed-set setting, the identities represented in the evaluation data are known to the system, allowing the experiment to measure how accurately the learned representations retrieve the correct individual. In the open-set setting, some identities are not present during training, requiring the system to distinguish known individuals from previously unseen ones. These two settings allow the project to evaluate both individual-level retrieval and the ability of the proposed representation to generalise beyond identities encountered during training.

Animal detection and cropping. Before extracting visual representations, each image is processed using MegaDetector [**?**]. The detector is used to localise the animal and produce a cropped image containing the relevant subject. This step is intended to reduce the influence of background information, such as vegetation, terrain, camera position, and lighting conditions, which may otherwise provide information correlated with an individual's identity.

The same detection and cropping procedure is applied before extracting the representations used in the experiments. This provides a more consistent input to the different descriptors and reduces the possibility that differences in

background content determine the observed performance differences between representations.

Global representation. Two wildlife-specific global descriptors are considered. The first is MiewID [**?**], which is used as a pre-trained representation for individual animal recognition. The second is MegaDescriptor [**?**], which is fine-tuned on the WildlifeReID-10k training data using an ArcFace-style angular-margin objective [**?**].

The purpose of considering both models is not to assume that one is universally superior, but to determine which global representation provides the strongest basis for the subsequent fusion experiments. Both representations are evaluated under the same retrieval protocol before the stronger candidate is selected for the main fusion experiments.

The fine-tuning procedure is performed using training data only. Validation data are used for model selection and parameter tuning, while the test data remain isolated until the final evaluation. This separation prevents information from the evaluation set from influencing the learned global representation.

Complementary representation. EVA02 [**?**] is used as a second visual representation. Unlike MiewID and MegaDescriptor, EVA02 is not specifically trained for wildlife individual re-identification. Instead, it provides a general visual representation learned from large-scale image data. It is therefore treated in this project as a complementary visual source rather than as a dedicated wildlife identity descriptor.

The purpose of including EVA02 is to test whether information contained in a general visual representation can provide additional discriminative information when combined with a wildlife-specific global descriptor and local keypoint evidence. Its contribution is evaluated through ablation experiments that compare the global representation alone with combinations of the global, EVA02, and local representations.

Local representation. Local visual information is obtained using ALIKED [**?**]. Unlike MiewID, MegaDescriptor, and EVA02, ALIKED is not treated as a fixed-length image embedding. It detects local keypoints and produces correspondences between two images. The resulting matching information provides evidence about whether fine-grained spatial patterns are shared between the two images.

This local information is particularly relevant to individual re-identification because animals of the same species may have similar overall morphology while differing in smaller visual details. Depending on the species, these details may include spot arrangements, stripe patterns, markings, scars, or other local structures. The local matching score is therefore treated as a complementary source of identity evidence rather than being concatenated directly with the global embeddings.

General-purpose baseline. DINOv2 [12] is included as a general-purpose baseline. Its embeddings are extracted without wildlife-specific fine-tuning and evaluated using the same retrieval framework as the wildlife-specific representations.

The purpose of this baseline is to establish whether the proposed wildlife-specific and fused representations provide an advantage over a strong general-purpose visual representation. DINOv2 does not form part of the proposed fusion itself; it is used as a separate comparison point.

Score calibration. The outputs of the different representations cannot be combined directly because they have different numerical meanings and scales. Global descriptors produce vector similarities, typically measured using cosine similarity, whereas ALIKED produces a keypoint-matching score. Directly averaging these raw values could therefore cause one representation to dominate the fusion simply because its scores have a different numerical range.

Following WildFusion [7], the raw similarity scores are calibrated before fusion. Positive and negative image pairs are sampled from the training or validation data, and the relationship between raw similarity scores and pairwise matching outcomes is learned using isotonic regression. PCHIP interpolation is then used to obtain a smooth monotonic calibration function.

The calibration parameters are estimated without using the final test set. This ensures that the test set remains unseen during calibration and provides an unbiased evaluation of the final system.

Multi-scale feature fusion. After calibration, the global, complementary, and local evidence is combined. The fusion is performed at the similarity-score level rather than by directly concatenating the outputs of the three models. This is necessary because ALIKED provides pairwise matching scores rather than a conventional fixed-dimensional embedding.

Let S_g denote the calibrated global similarity, S_c the calibrated EVA02 similarity, and S_l the calibrated ALIKED similarity. The proposed fusion can be expressed as

$$S_{\text{fusion}} = w_g S_g + w_c S_c + w_l S_l,$$

where

$$w_g + w_c + w_l = 1$$

and each weight is non-negative.

The weights are selected using validation data and are not estimated using the final test set. Different configurations are evaluated, including the global representation alone, global plus local fusion, global plus complementary fusion, and the complete global-complementary-local fusion. These comparisons provide an ablation-based test of whether each representation contributes useful information beyond the others.

The global descriptor used in the final fusion is selected based on the comparison between MiewID and fine-tuned MegaDescriptor. This prevents the choice of global descriptor from being assumed in advance and allows the experiments to establish which representation provides the stronger foundation for multi-scale fusion on WildlifeReID-10k.

PCA-based compression. The proposed fused representation is subsequently examined under different dimensionality constraints. PCA [**?**] is applied to the

fused representation using training data, and several target dimensionalities are evaluated.

The purpose of this experiment is to determine how much dimensionality can be removed while preserving individual-level discrimination. The unreduced representation is used as the reference condition, and retrieval performance at progressively smaller dimensionalities is compared with this reference.

Compression is therefore treated as an experimental variable rather than simply as an engineering optimisation. The results will show whether the fused representation contains redundant information and identify the point at which dimensionality reduction begins to negatively affect individual re-identification performance.

Similarity retrieval. The resulting representation is used for similarity-based retrieval. Given a query image, the system compares its representation with representations stored in the gallery and ranks candidate images according to their similarity to the query.

Locality-sensitive hashing (LSH) [**?**] may be used to accelerate similarity search for the large gallery. However, the primary research outcome remains retrieval accuracy rather than indexing performance. Retrieval latency and memory requirements are recorded as secondary measures to characterise the practical cost of the proposed representation.

Closed-set evaluation. In the closed-set experiment, the correct individual is assumed to be represented within the known identity set. For each query image, the gallery images are ranked according to the final similarity score, and the position of the correct identity is used to evaluate retrieval performance.

The fused representation is compared against the individual descriptors and relevant ablation configurations. This experiment directly addresses whether combining complementary visual evidence improves individual-level retrieval compared with relying on a single representation.

Open-set evaluation. The open-set experiment evaluates whether the proposed representation can handle individuals that were not observed during training. Rather than forcing every query to match an existing identity, the system applies a decision threshold to determine whether the strongest gallery match is sufficiently similar to be considered a known individual.

Let $S_{\max}$ denote the similarity score of the strongest gallery match. The decision is defined as

$$\hat{y} = \begin{cases} \text{known identity}, & S_{\max} \geq \tau, \\ \text{unknown individual}, & S_{\max} < \tau, \end{cases}$$

where τ is the open-set decision threshold.

The threshold is selected using validation data and is not determined using the final test set. Where supported by the benchmark protocol, thresholds are calibrated separately by species to account for differences in visual similarity between individuals of different species. The final evaluation is then performed on the held-out test data.

Performance is reported using the evaluation measures specified by WildlifeReID-10k, including the closed-set and open-set metrics used by the benchmark. This allows the proposed method to be assessed under the same evaluation framework as existing WildlifeReID-10k results.

Ablation and comparative evaluation. The experiments are structured to determine which parts of the proposed methodology contribute to performance. The main comparisons include MiewID, fine-tuned MegaDescriptor, EVA02, ALIKED, and DINOv2 as individual or baseline representations, followed by combinations of global, complementary, and local evidence.

The complete fusion is then compared with reduced fusion configurations. These comparisons allow the project to determine whether improvements arise from the fusion itself or simply from using a stronger individual descriptor. The compression experiments further determine whether the observed improvements remain after reducing the dimensionality of the fused representation.

The final results are reported separately for closed-set and open-set evaluation and, where supported by the benchmark, across the different species and source datasets. This makes it possible to assess not only overall retrieval performance but also whether the observed benefit is consistent across the heterogeneous data represented in WildlifeReID-10k.

The research methodology described above represents only the core experimental pipeline. Any inference API, web application, or other software interface developed for demonstration or course requirements is treated as a separate implementation deliverable. These components are not part of the proposed research pipeline and are not used to define the scientific methodology or the experimental evaluation.

5 Justification and Conclusion

The proposed study addresses the research gap identified in Section 3: although calibrated fusion of global, local, and complementary visual representations has shown promising results for wildlife Re-ID, it remains unclear whether these representations provide consistent additional information when evaluated together at the scale and diversity of WildlifeReID-10k. Existing studies have demonstrated the value of combining different sources of visual evidence, but the extent to which this benefit transfers to a large benchmark containing many species, source datasets, and unseen individuals remains insufficiently established. This project therefore evaluates the proposed fusion directly on WildlifeReID-10k using its established evaluation protocols.

The choice of representations is motivated by the different types of information that they provide. MiewID and MegaDescriptor provide wildlife-specific global representations that capture broader visual characteristics associated with individual animals. EVA02 provides a general visual representation that may contain information not captured by the wildlife-specific descriptor, while ALIKED provides local correspondence information that can capture fine-grained spatial patterns. Considering these representations together allows the project to test

whether different forms of visual evidence are complementary at the individual level. The direct comparison between MiewID and fine-tuned MegaDescriptor also prevents the choice of global descriptor from being assumed in advance.

Calibration is an important part of the proposed approach because the representations do not produce directly comparable similarity scores. Applying calibration before fusion allows the contribution of each representation to be combined on a common scale. The ablation experiments then provide a direct way of determining whether the observed performance improvement is actually attributable to complementary information from multiple representations rather than simply to the strength of one individual descriptor.

The evaluation under both closed-set and open-set conditions addresses two different aspects of individual re-identification. Closed-set evaluation measures the ability to retrieve the correct individual when the relevant identity is represented in the known population. Open-set evaluation extends this problem by requiring the system to recognise when a query does not correspond to a known individual. This distinction is important for wildlife monitoring because populations are not static: individuals may enter a monitored population or appear in the data without having been observed previously. Evaluating both settings therefore provides a stronger assessment of whether the proposed representation can support realistic individual-level re-identification.

The study also evaluates the proposed representation across the heterogeneous data contained in WildlifeReID-10k. The benchmark combines multiple source datasets and species that differ in appearance, image conditions, and the number of images available for each identity. Reporting performance across these conditions allows the project to determine whether the benefit of fusion is consistent or whether it depends strongly on particular species or source collections. This is important because a method that performs well only on a subset of the benchmark would provide weaker evidence for general multi-species wildlife re-identification.

Finally, the compression experiment investigates how much of the fused representation is actually necessary for individual-level discrimination. PCA is used to produce representations of different dimensionalities, and retrieval performance is compared with the unreduced representation. This experiment does not assume that the highest-variance dimensions are necessarily the most useful for identity recognition. Instead, it empirically determines whether dimensionality can be reduced while maintaining useful individual-level information. Runtime and memory measurements are included as secondary indicators of practical efficiency, while retrieval accuracy remains the primary research outcome.

Taken together, these experiments provide a structured evaluation of multi-scale calibrated feature fusion for wildlife individual Re-ID. The study does not assume that fusion will necessarily improve performance. Instead, it tests whether global, complementary, and local representations provide measurable additional information, whether that benefit extends to unseen individuals and diverse species, and how much of the resulting representation can be removed before performance begins to degrade. Comparisons with individual descriptors

and DINOv2 provide appropriate reference points for interpreting any observed improvement.

The expected contribution of the project is therefore empirical rather than a claim of a universally superior re-identification system. The study aims to provide evidence about the usefulness, generalisability, and compression tolerance of calibrated multi-scale feature fusion on a large, heterogeneous wildlife Re-ID benchmark. If the fused representation consistently outperforms the individual representations under the benchmark's evaluation protocols, this would support the hypothesis that combining complementary visual evidence is beneficial for individual wildlife re-identification. Conversely, limited or inconsistent improvements would provide evidence about the conditions under which such fusion does not provide sufficient additional information to justify its complexity.

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